

TUTORIAL – 3

Q1. In the Playfair Cipher, suppose that the keyword is KEYWORD, how do you encrypt SECURE ELECTRONIC COMMERCE?

Q2. With the Simplified DES, encrypt 10011101 with key 0110101011.

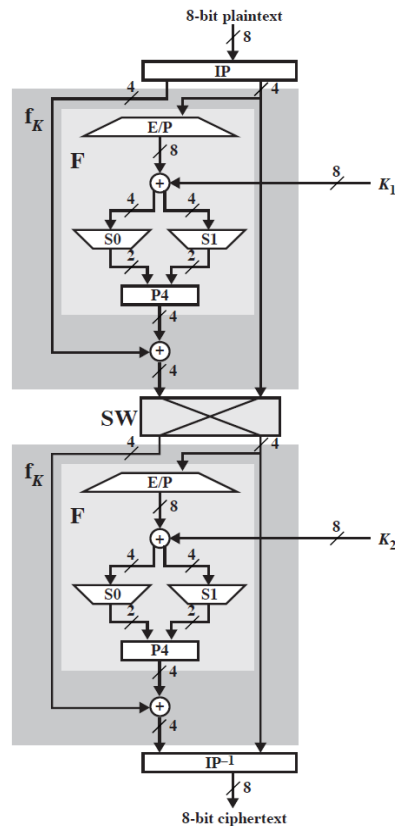
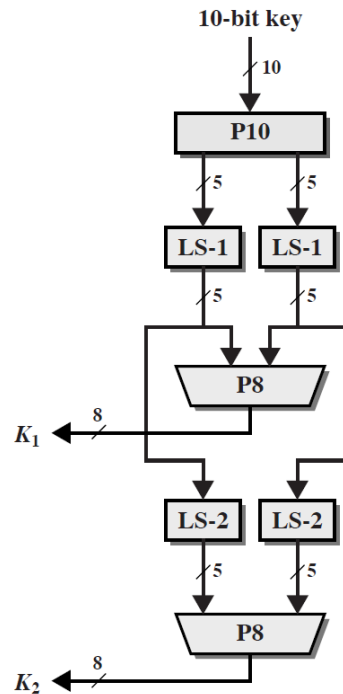


Figure C.3 Simplified DES Encryption Detail



P10									
3	5	2	7	4	10	1	9	8	6

P8									
6	3	7	4	8	5	10	9		

IP							
2	6	3	1	4	8	5	7

IP ⁻¹							
4	1	3	5	7	2	8	6

P4			
2	4	3	1

E/P							
4	1	2	3	2	3	4	1

$$S0 = \begin{matrix} 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 1 \\ 2 & 0 & 2 & 1 \\ 3 & 3 & 1 & 3 \end{matrix} \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix}$$

$$S1 = \begin{matrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{matrix} \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix}$$



[Click here to watch on YouTube](#)

Playfair Cipher

Keywords:

No repeating Letters

Table:

Remove J and Insert the keyword in the 5 X 5 Table

Complete Filling it with all Unique Alphabets remaining

Message:

Split into Pairs, Odd and Duplicate Letters Paired with X

Ignore Spaces

SECRET MESSAGE
SE CR ET ME SX SA GE

Insert each pair into a separate table

1. IF in same COLUMN

- Move each letter down ONE
- Upon reaching end of table, wrap around

2. IF in same ROW

- Move each letter right ONE
- Upon reaching end of table, wrap around

3. IF it forms a rectangle

- Swap the letters with the ones on the end of the rectangle

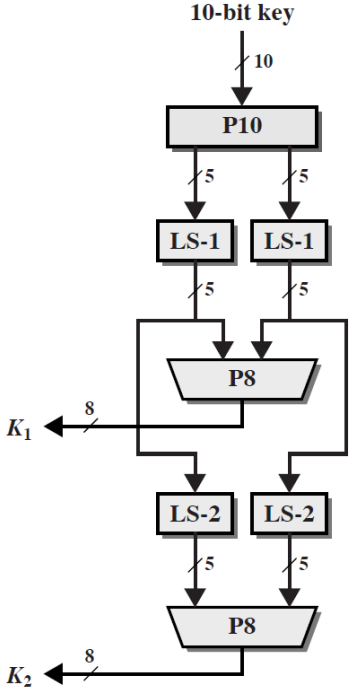
• Original Text
SECRET MESSAGE
SECRETMESSAGE
SE CR ET ME SX SA GE

• Cipher Text
NO RD KU NK QZ PC ND
NORDKUNKQZPCND

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	S
T	U	V	X	Z

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	S
T	U	V	X	Z

DES Obtaining K1 and K2



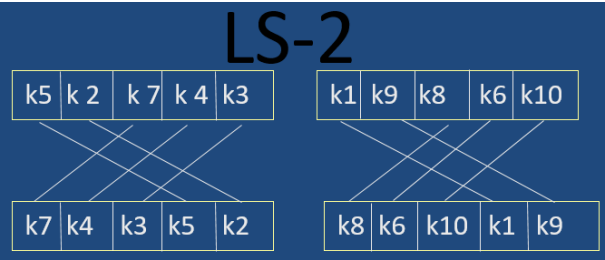
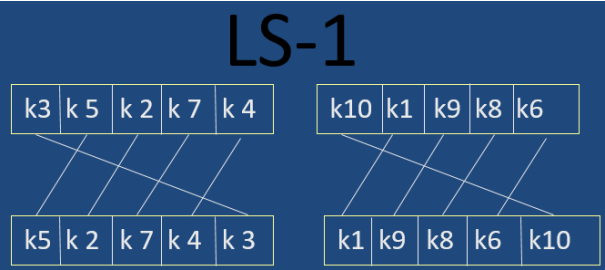
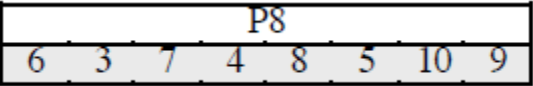
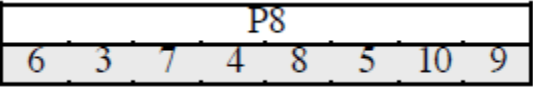
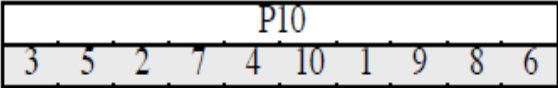
0110101011
1111010100

11101 01001

01100110

10111 00101

01011110



DES Encryption

Text = **10011101** K1 = **01100110** K2 = **01011110**

IP							
2	6	3	1	4	8	5	7

E/P							
4	1	2	3	2	3	4	1

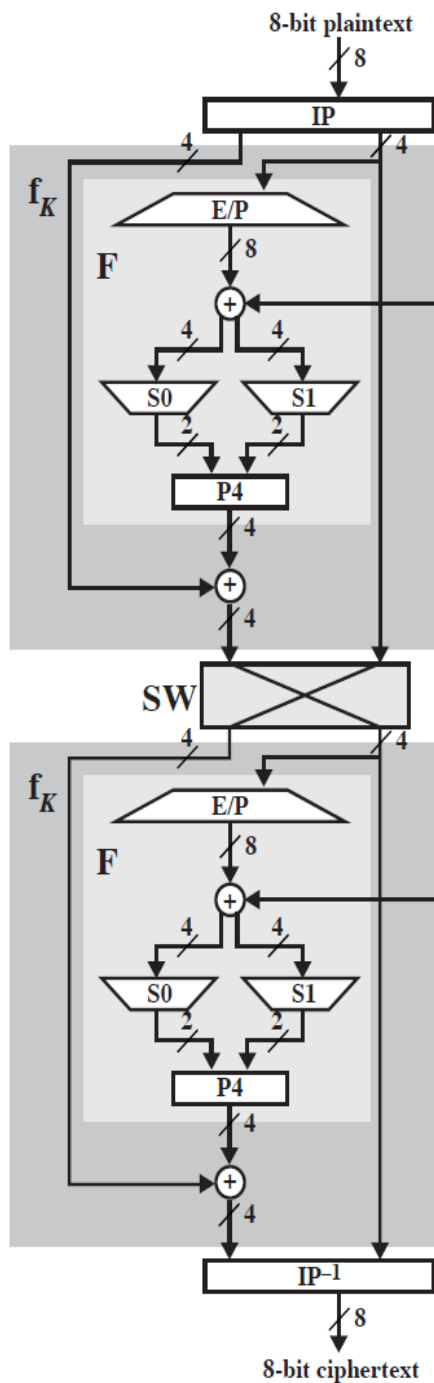
K1 = **01100110**

S0(**0000**)=1=01 S1(**1011**)=1=01
S0(**0001**)=3=11 S1(**0110**)=3=11

P4			
2	4	3	1

	0	1	2	3
0	1	0	3	2
1	3	2	1	0
2	0	2	1	3
3	3	1	3	2

	0	1	2	3
0	0	1	2	3
1	2	0	1	3
2	3	0	1	0
3	2	1	0	3



10011101

0101 1110

01111101

+01100110

0001 1011

11 01

1101

+ **0101**

1000

1110 1000

01000001

+01011110

0001 1111

11 11

1111

+ **1110**

0001

0001 1000

10010000

K2 = **01011110**

E/P							
4	1	2	3	2	3	4	1

S0(**0000**)=1=01 S1(**0010**)=1=01
S0(**0001**)=3=11 S1(**0110**)=3=11

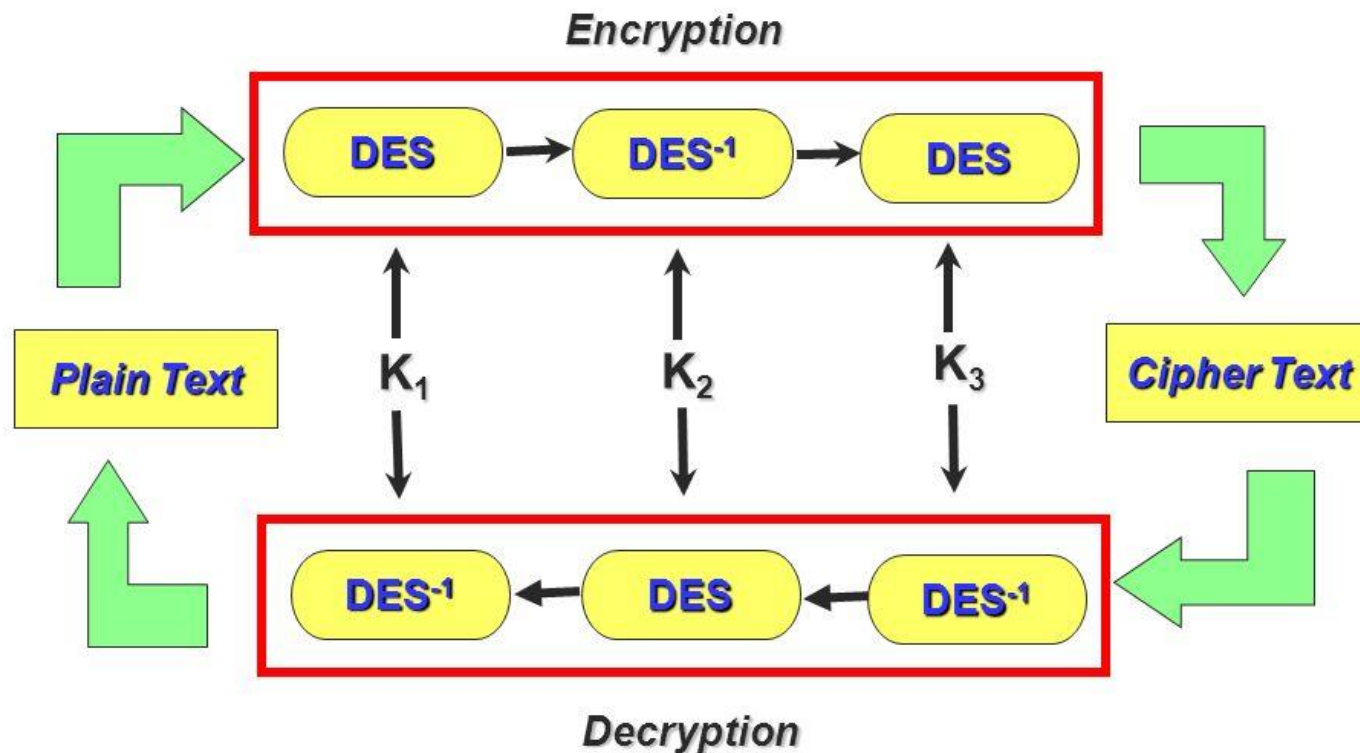
P4			
2	4	3	1

IP ⁻¹							
4	1	3	5	7	2	8	6

D ata
E ncryption
S andard

Next

TRIPLE DATA ENCRYPTION STANDARD (3DES)



Use 3 keys and 3 executions

Effective key length of 168 bits

$$C = E_{K3}[D_{K2}[E_{K1}[P]]] \quad \text{Encryption:}$$

$$P = D_{K1}[E_{K2}[D_{K3}[C]]] \quad \text{Decryption}$$

C = ciphertext, P = Plaintext

EK[X] = encryption of X using key K

DK[Y] = decryption of Y using key K

Key Option # 1

K_1, K_2, K_3 are Different

Strongest key Bits $56 \times 3 = 168$

Key Option # 1

$K_1, K_2, K_3 = K_1$, 2 Different K

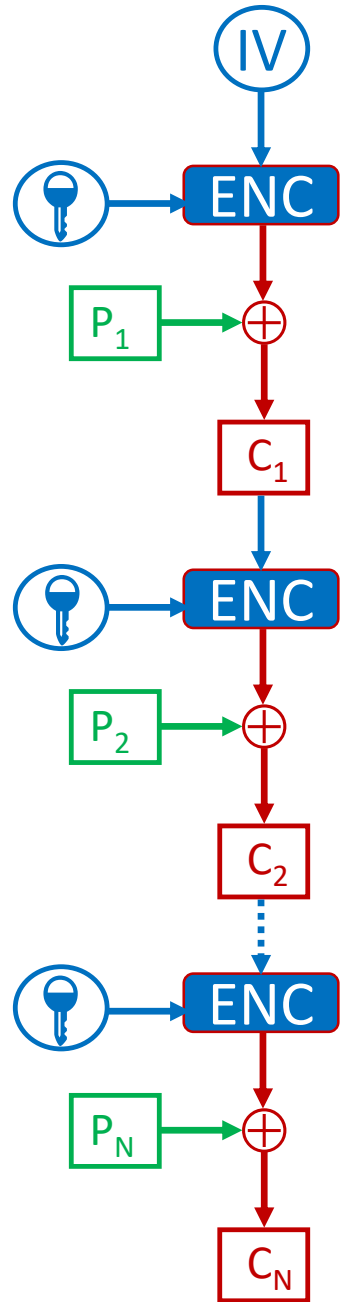
Reasonable key Bits $56 \times 2 = 112$

Key Option # 1

$K_1, K_2 = K_1, K_3 = K_1$, All Identical

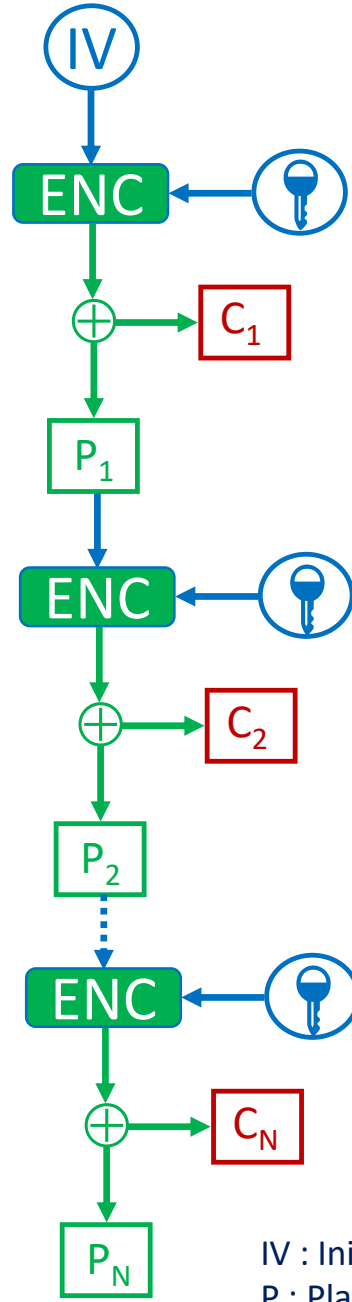
Reasonable key Bits $56 \times 1 = 56$

Encryption



Cipher Feedback [CFB] DES

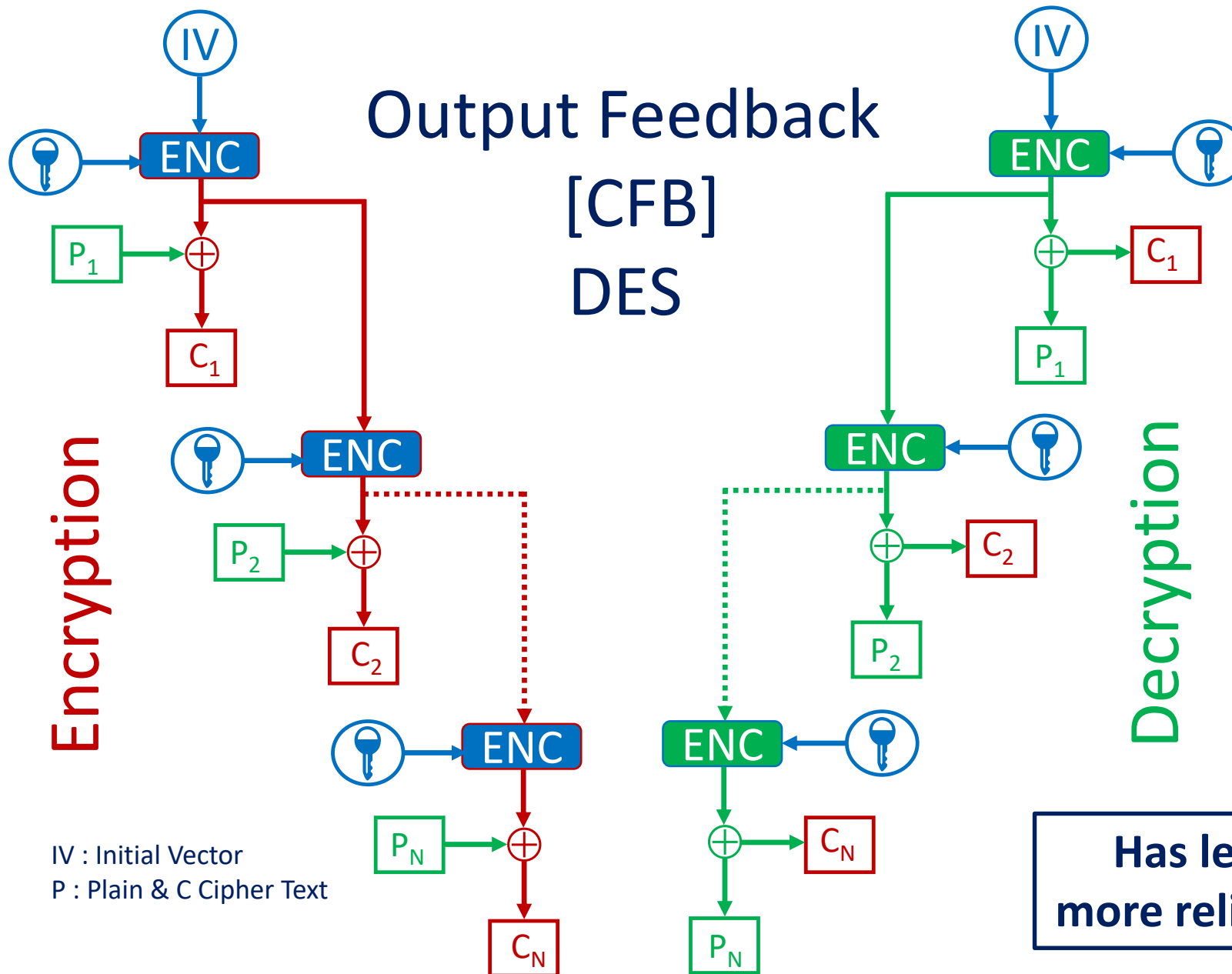
Decryption



IV : Initial Vector
P : Plain & C Cipher Text

If a bit of P is damaged,
dependent C and all coming
C will be damaged

Data loss due to shift register
makes cryptanalysis difficult.



If a bit of P is damaged,
Only one bit C will
be damaged

Has less error propagation,
more reliable in communication.